Multimodal Probabilistic Photovoltaic Forecasting: A Matched-Checkpoint Study of DDIM Sampling

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***Abstract*—Fast sampling is attractive for probabilistic photovoltaic (PV) forecasting, but a low aggregate error does not establish reliable prediction intervals. We examine this trade-off by sampling one conditional multimodal diffusion checkpoint with a 1000-step denoising diffusion probabilistic model (DDPM) sampler and ten-step denoising diffusion implicit model (DDIM) samplers. The network jointly denoises future sky images and PV power while retaining the observed history. On 6627 windows from 20 SKIPP'D test days, ten-step DDIM achieves an empirical continuous ranked probability score (CRPS) of 1.144 kW at the furthest target, with 41.4% skill over the implemented smart-persistence reference. A separate matched comparison uses 128 windows and ten ensemble members. There, selected DDIM reduces latency from 21.17 to 0.21 s per window and CRPS from 0.982 to 0.886 kW, while central empirical 90% interval coverage falls from 87.5% to 64.8%. The paired day-bootstrap CRPS difference is -0.096 kW (95% interval $[-0.181, -0.022]$). These results establish a speed–coverage trade-off for the evaluated ensembles, not its cause: finite ensemble size, random attention offsets, and the absence of intermediate-step ablations limit attribution. Larger-ensemble verification and independent recalibration remain necessary before interval-based deployment.**

***Index Terms*—photovoltaic forecasting, diffusion models, DDIM, sky images, probabilistic forecasting, ensemble verification**

I. Introduction

Short-term PV forecasting is difficult when clouds alter the radiation reaching a plant faster than power-history extrapolation can anticipate. Ground-based sky images provide complementary information about local cloud conditions, motivating models that combine images with recent power measurements [1], [2]. For applications that use prediction intervals, forecasting accuracy and uncertainty reliability must be assessed separately: sharp intervals are useful only when their coverage is appropriate [3].

Conditional diffusion offers a way to generate multiple plausible outcomes from the same history. PV-MM-Diffusion [4] adapts the coupled denoisers and cross-modal interaction of MM-Diffusion [5] to sky images and PV power. Repeated ancestral sampling, however, makes ensemble generation expensive. DDIM [6] can sample a DDPM-trained network [7] on fewer diffusion steps without retraining it. The practical question is whether that computational saving preserves the properties of the resulting forecast ensemble.

This paper examines that question within one simplified multimodal model. Its contribution is an empirical evaluation, rather than a new diffusion objective or sampler. We report a matched-checkpoint comparison of latency, point errors, CRPS, interval width, and coverage; a validation-only sweep of DDIM stochasticity; and an analysis of what these measurements can and cannot establish. The central finding is that the tested ten-step configuration is much faster and has lower endpoint empirical CRPS, yet its delivered intervals miss substantially more observations. The distinction between a finite forecast ensemble and its underlying generative distribution is essential to interpreting this result.

II. Related Work

Sky-image forecasting ranges from direct image–power regression [1] to generative approaches that first produce future sky scenarios. SkyGPT [8] and the two-stage framework of Pan and Lim [9] use such scenarios for probabilistic PV prediction. PV-MM-Diffusion [4] instead generates both modalities jointly. Our model follows this latter formulation, but removes temporal layers from the sky branch and pools image features before cross-modal attention. It is therefore not an exact reproduction of PV-MM-Diffusion.

DDIM [6], DPM-Solver [10], and DPM-Solver++ [11] establish that diffusion sampling can be accelerated. Applying DDIM to a conditional network does not itself constitute a new sampling algorithm. Here, the narrower contribution is its evaluation under a common forecasting checkpoint and test subset. Proper scoring rules summarize predictive performance [12], whereas coverage diagnostics reveal particular calibration deficiencies [3]. Moreover, empirical ensemble scores depend on ensemble size and need not rank latent forecast distributions in the same way as fair ensemble scores [13]. These distinctions motivate reporting both aggregate scores and interval diagnostics.

III. Method

A. Forecast target and conditional diffusion

A window contains $L = 16$ frames at two-minute spacing. Let $P \in \mathbb{R}^{16}$ denote PV power and $S \in \mathbb{R}^{16\times3\times64\times64}$ denote sky images. The timestamp convention assigns context indices 0:7 to $t-15, t-13, \ldots, t-1$ minutes and target indices 8:15 to $t+1, t+3, \ldots, t+15$ minutes. Thus t is a reference

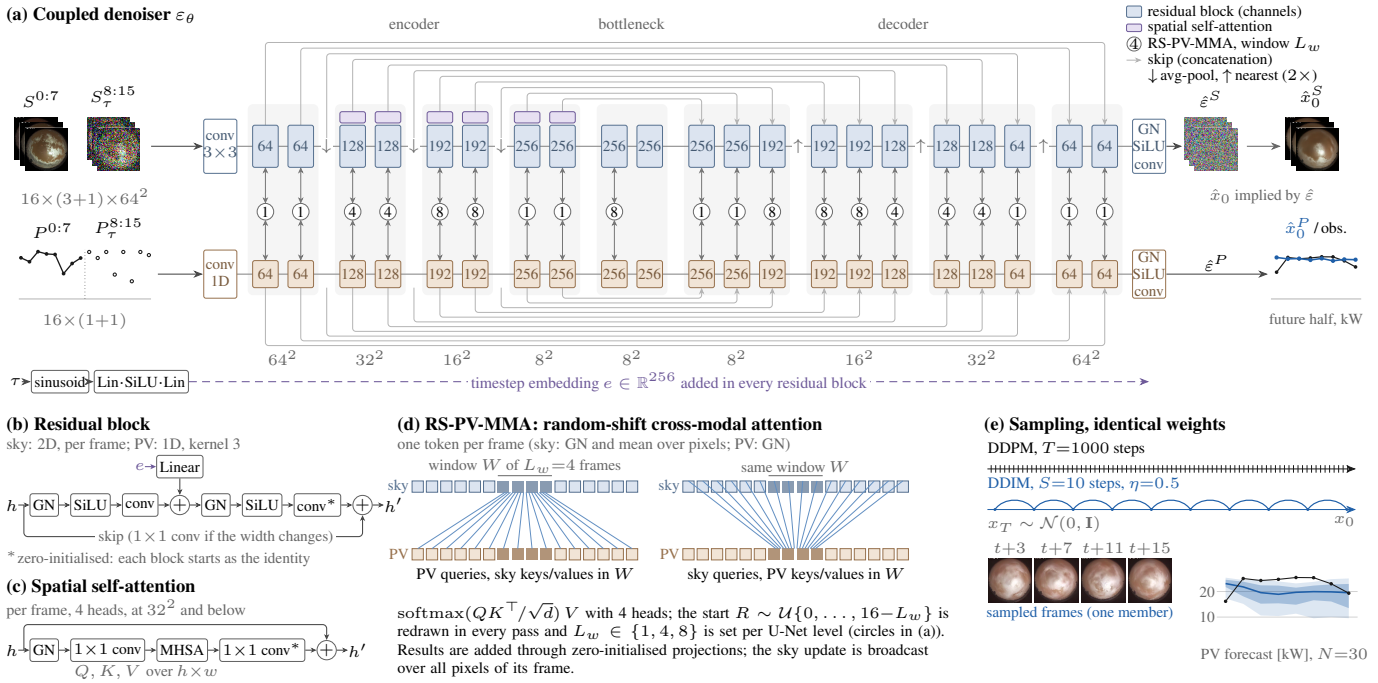


Fig. 1. The 29.1-million-parameter coupled denoiser. (a)–(d) Sky and PV branches, residual blocks, spatial attention, and pooled random-shift cross-modal attention. (e) Sampling illustration for the same checkpoint. The window offset is redrawn during forward passes; consequently $\eta = 0$ does not remove all implementation-level randomness. Generated sky frames are illustrative outputs, not validated cloud forecasts.

timestamp, not the time of the latest input. The plotted leads $1, 3, \dots, 15$ correspond to elapsed leads $2, 4, \dots, 16$ minutes from the latest observation. All results use this reference-time convention; the endpoint is denoted $t + 15$ below.

The model represents $p_\theta(S^{8:15}, P^{8:15} \mid S^{0:7}, P^{0:7})$. Only future segments are corrupted. For either modality, with diffusion index $\tau \in \{1, \dots, T\}$,

$$x_\tau = \sqrt{\bar{\alpha}_\tau}\, x_0 + \sqrt{1 - \bar{\alpha}_\tau}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I), \tag{1}$$

where $\alpha_\tau = 1 - \beta_\tau$ and $\bar{\alpha}_\tau = \prod_{s=1}^{\tau} \alpha_s$. Clean context and noisy future are concatenated along time, with an observation-mask channel identifying available history.

B. Coupled denoiser and masked-history training

The sky branch is a per-frame U-Net with channel widths 64, 128, 192, and 256, two residual blocks per encoder level, and spatial self-attention at 32×32 pixels and below. The PV branch uses aligned one-dimensional residual blocks. Cross-modal attention exchanges one pooled sky token and one PV token per frame within windows of $L_w \in \{1, 4, 8\}$ frames. Window starts are drawn from $\{0, \dots, 16 - L_w\}$ on each forward pass, and attention updates use zero-initialized projections. Unlike the parent architecture [4], [5], the sky branch has no explicit temporal convolutions or temporal attention (Fig. 1).

Training samples use both histories (SP), sky history only (SI), or PV history only (PV), each with probability $1/3$. An unavailable history and its mask are zeroed. The future-segment loss is

$$\mathcal{L} = \mathbb{E}_{m,\tau,\varepsilon,R}\left[\overline{(\hat{\varepsilon}_\theta^P - \varepsilon^P)^2} + \mathbf{1}[m \neq \mathrm{PV}]\,\overline{(\hat{\varepsilon}_\theta^S - \varepsilon^S)^2}\right], \tag{2}$$

where each bar averages within its modality and R denotes attention-window randomness. The image loss is absent in PV-only mode. This supports history-masking evaluation, but does not establish that image generation is useful in every mode. All sampler comparisons use the same trained weights.

C. Sampling and clipping consistency

For predicted noise $\hat{\varepsilon}_\theta$, the clean estimate is

$$\hat{x}_0 = \frac{x_\tau - \sqrt{1 - \bar{\alpha}_\tau}\hat{\varepsilon}_\theta}{\sqrt{\bar{\alpha}_\tau}}, \qquad \tilde{x}_0 = \mathrm{clip}(\hat{x}_0, -1, 1). \tag{3}$$

Let $p < \tau$ be the next, less noisy sampling index. DDIM traverses a selected subsequence in decreasing order, using $\bar{\alpha}_0 = 1$ at the terminal state:

$$\sigma_{\tau,p} = \eta\sqrt{\frac{1 - \bar{\alpha}_p}{1 - \bar{\alpha}_\tau}}\sqrt{1 - \frac{\bar{\alpha}_\tau}{\bar{\alpha}_p}}, \tag{4}$$

$$\tilde{\varepsilon} = \frac{x_\tau - \sqrt{\bar{\alpha}_\tau}\tilde{x}_0}{\sqrt{1 - \bar{\alpha}_\tau}}, \tag{5}$$

$$x_p = \sqrt{\bar{\alpha}_p}\tilde{x}_0 + \sqrt{1 - \bar{\alpha}_p - \sigma_{\tau,p}^2}\,\tilde{\varepsilon} + \sigma_{\tau,p} z, \tag{6}$$

with fresh standard Gaussian z for nonterminal stochastic updates. Recomputing the direction from the clipped estimate preserves consistency between the terms; this is established implementation practice, not a new contribution [14].

With every diffusion index retained, $p = \tau - 1$ and $\eta = 1$ give the fixed-posterior DDPM variance $\tilde{\beta}_\tau = \beta_\tau(1 - \bar{\alpha}_{\tau-1})/(1 - \bar{\alpha}_\tau)$ and the same posterior-mean update when clipping conventions match. This is an algebraic equivalence, not a claim that arbitrary software implementations have identical random-number consumption. At $\eta = 0$, the DDIM update adds no Gaussian noise after initialization. A deterministic mapping from initial noise additionally requires fixed network behavior; the random attention offsets used here do not satisfy that condition automatically.

Sampling begins with Gaussian noise in both future segments. Observed history remains fixed, and both modalities are updated at each step. With M members and K sampling steps, cost is K denoiser evaluations per member, or MK member-evaluations per window. Batched forward calls need not equal MK.

IV. Experimental Setup

A. Data and training

SKIPP'D provides synchronized sky images and PV measurements from Stanford for 2017–2019 [2]. The benchmark contains 64×64 images and one-minute power measurements. Preprocessing keeps complete 16-frame windows at exact two-minute spacing within one day, holds out ten development days for validation, and uses the 20 benchmark test days. The resulting training, validation, and test counts are 161059, 3604, and 6627 windows. Whole-day separation prevents overlapping within-day windows from entering different splits. The within-window spacing does not determine the window-start stride.

Power normalization uses the training 99.9th percentile, 28.4 kW, as the upper scale. Sampling clips normalized outputs to $[-1, 1]$. This percentile-based cap is not the physical capacity and can affect upper-tail predictions; its empirical effect has not been isolated.

Training uses $T = 1000$, linearly spaced β_τ from 10^{-4} to 0.02, and 10^5 AdamW [15] updates with batch size 32. The configuration uses learning rate 10^{-4}, warm-up followed by cosine decay, gradient clipping, bfloat16, and exponential moving-average weights with decay 0.9999. All evaluations use the validation-selected checkpoint at step 92000. Training time is 4.7 h on one AMD Instinct MI300X with compiled execution and fused attention. This is one training run, not a multi-seed estimate.

B. Selection, test protocols, and references

A validation sweep evaluates five η values on 512 windows with $K = 10$, a uniform timestep subsequence, and $M = 10$. The smallest endpoint Winkler score selects $\eta = 0.5$; the test set does not select η. Two distinct test protocols then apply. The full test uses $M = 30$ on all 6627 windows for each input mode. The matched sampler comparison uses $M = 10$ on 128 windows drawn across all 20 test days. The latter compares 1000-step DDPM with ten-step DDIM at $\eta = 0.5$ and $\eta = 0$. Its conclusions apply to that subset, not to an unreported full-test DDPM run.

The 95% intervals use 10000 day-block bootstrap resamples, paired when estimating sampler differences. They describe sampled-day variability conditional on the trained checkpoint and evaluated random seed. They do not capture training variability or sensitivity to alternative sampling seeds. Sharing a seed is also weaker than explicitly sharing initial noise and separating random-number streams for Gaussian innovations and attention offsets.

The deterministic references are persistence and smart persistence. The latter holds the latest clear-sky index constant and extrapolates using a solar-geometry power surrogate [9]: a 1000 W/m^2 beam, tilt $22.5°$, azimuth $195°$, and effective area 24.98 m^2. Reported skill is relative to this implementation. These skills are implementation-specific; reproducing them requires identical as-of timestamps, low-sun handling, and solar-time conventions.

C. Metrics and finite-ensemble interpretation

For M members x_i and observation y, the exact CRPS of their empirical distribution is

$$\mathrm{CRPS}_M = \frac{1}{M}\sum_i |x_i - y| - \frac{1}{2M^2}\sum_{i,j} |x_i - x_j|. \quad (7)$$

It equals absolute error for a point forecast. For empirical 5th and 95th percentiles l, u, the 90% Winkler interval score is

$$\mathrm{WS} = (u-l) + \frac{2}{0.1}\left[(l-y)\mathbf{1}[y < l] + (y-u)\mathbf{1}[y > u]\right]. \quad (8)$$

PICP is the fraction of observations inside the interval. Width is the mean $u - l$; MAE and RMSE use the ensemble median. CRPS skill is $100(1 - \mathrm{CRPS}_M / \mathrm{MAE}_{\mathrm{ref}})\%$. CRPS is a proper distributional score, not merely a sharpness measure [12]; lower average CRPS nevertheless does not imply correct coverage at a specific nominal level.

Equation (7) evaluates the delivered finite ensemble. For independent draws from a latent forecast distribution, its expected upward bias relative to that distribution's CRPS is $\mathbb{E}|X - X'|/(2M)$. Replacing M^2 with $M(M-1)$ in the pairwise term gives the fair estimator under those assumptions [13]. The results use the empirical score throughout; fair-score sensitivity cannot be reconstructed from aggregate tables alone.

Small ensembles also limit tail-quantile estimation. For M independent draws and an independent observation from the same continuous distribution, exchangeability gives

$$\Pr\{X_{(1)} \le Y \le X_{(M)}\} = \frac{M-1}{M+1}. \quad (9)$$

For $M = 10$, even this full ensemble range has expected coverage $9/11 \approx 81.8\%$; interpolated interior quantiles are narrower. This analytical benchmark is not a correction to the observed PICP, and does not bound a finite test sample's realized coverage. It shows why a 90% label alone is insufficient to diagnose calibration of the underlying distribution. Quantile conventions and larger-ensemble checks are required.

TABLE I
VALIDATION SWEEP: 512 WINDOWS, $K = 10$, $M = 10$, ENDPOINT $t + 15$. ERRORS, WS, AND WIDTH ARE IN KW.

η	CRPS↓	WS↓	MAE	RMSE	PICP	Width
0.00	0.3784	4.518	0.477	1.210	0.703	1.157
0.25	0.3674	4.458	0.453	1.152	0.693	1.092
0.50	0.3614	**4.421**	0.450	1.127	0.697	1.033
0.75	**0.3576**	4.587	0.444	1.096	0.672	0.934
1.00	0.3583	4.840	0.437	1.086	0.598	0.760

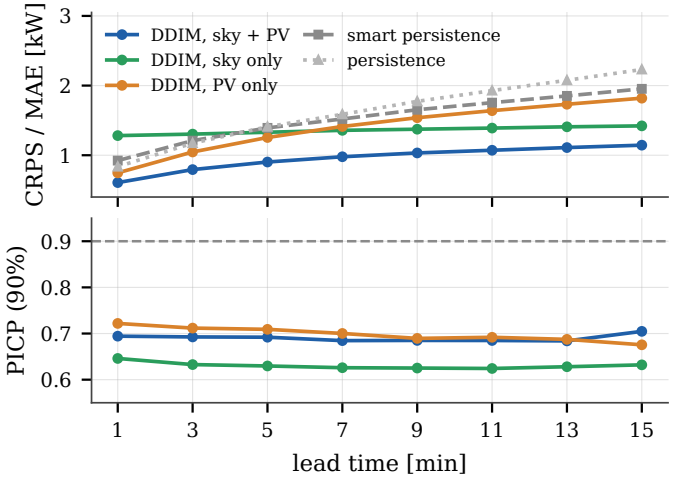


Fig. 2. Full-test empirical CRPS (MAE for point references) and empirical 90% coverage for the three history modes, $M = 30$. Lead labels are relative to t; elapsed time from the last input is one minute longer.

V. RESULTS

A. Validation selection

In Table I, width decreases from 1.157 to 0.760 kW as η rises from zero to one. The lowest CRPS occurs at $\eta = 0.75$, whereas the lowest WS occurs at $\eta = 0.5$. None of the five tested values produces empirical 90% coverage. These are validation observations, not a monotonicity result for DDIM in general. The modest differences between neighboring settings have not been assessed across repeated seeds.

B. Full-test performance and missing histories

With $K = 10$, $\eta = 0.5$, and $M = 30$, SP reaches endpoint CRPS $1.144\,\mathrm{kW}$ (95% interval $[0.66, 1.67]$), WS $11.80\,\mathrm{kW}$, and smart-persistence skill 41.4% (interval $[35.6, 51.1]\%$). Skill over persistence is 48.7%. Latency is 0.53 s per 30-member window. Endpoint CRPS is substantially higher than the validation-sweep value of 0.3614 kW; differences in window composition and ensemble size have not been isolated.

Across plotted leads, SP CRPS increases from about 0.61 to 1.14 kW, while coverage remains between 0.68 and 0.71 (Fig. 2). In the descriptive split by mean absolute one-minute power change, ten steadier days have endpoint CRPS 0.10 kW and PICP 0.87, compared with 2.20 kW and 0.54 on ten more variable days. This post-hoc stratification is descriptive, not a separately validated regime classifier.

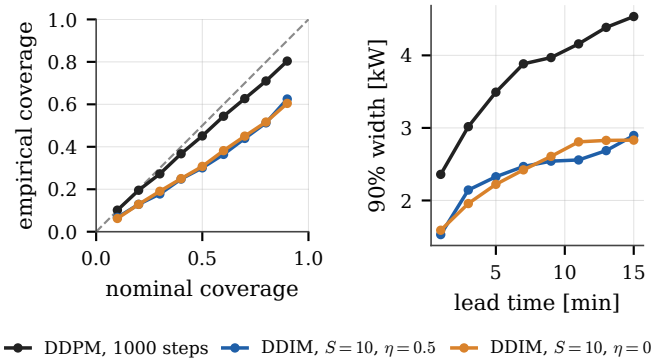


Fig. 3. Matched-subset diagnostics, $M = 10$. Left: coverage pooled over leads. Right: mean empirical 90% interval width. The nominal labels refer to finite-ensemble quantiles, not a proven population-coverage guarantee.

Removing PV history raises endpoint CRPS to 1.42 kW (skill 27.2%, PICP 0.63); removing sky history raises it to 1.82 kW (skill 6.9%). These masking results indicate complementary information within this trained model. They do not show that it outperforms separately optimized unimodal models or is robust to all real sensor failures.

C. Matched-checkpoint sampler comparison

Table II separates the sampler experiment from full-test reporting. Ten-step DDIM reduces member-evaluations by exactly $100\times$; the rounded latency values correspond to approximately $100\times$ wall-clock acceleration in this setting. This is not a measured reduction in energy use or deployment cost. Endpoint CRPS decreases by $0.096\,\mathrm{kW}$, with paired difference interval $[-0.181, -0.022]\,\mathrm{kW}$. Averaged over all leads, the difference is $-0.027\,\mathrm{kW}$ with interval $[-0.097, 0.037]\,\mathrm{kW}$. The latter does not establish either a difference or equivalence.

Selected DDIM produces intervals about 36% narrower than DDPM (2.895 versus 4.535 kW), with PICP 0.648 versus 0.875. The coverage difference is -0.227 (95% interval $[-0.297, -0.154]$). WS is higher for DDIM by about 1.05 kW, but its difference interval $[-0.46, 2.81]\,\mathrm{kW}$ includes zero. Figure 3 shows the associated coverage curves and interval widths.

DDIM at $\eta = 0$ has PICP 0.664 and width 2.832 kW. In particular, the selected $\eta = 0.5$ test ensemble is slightly wider, not narrower, than the $\eta = 0$ ensemble. Thus the monotonic validation pattern does not extend even to these two test settings. Both short configurations deliver lower coverage than DDPM, but the comparison changes step count and stochasticity together and does not identify the cause.

Published SKIPP'D-based results are not ranked against these numbers. For example, the Pan–Lim comparison [9] uses a five-cloudy-day evaluation rather than this 20-day benchmark protocol. Differences in weather composition, horizon conventions, ensemble sizes, and reference implementations prevent a state-of-the-art conclusion from cross-paper score tables.

TABLE II

MATCHED COMPARISON: 128 WINDOWS OVER 20 TEST DAYS, $M = 10$, ENDPOINT $t + 15$. ERRORS, WS, AND WIDTH ARE IN KW. NFE IS PER MEMBER; ME IS MEMBER-EVALUATIONS PER WINDOW, NOT NECESSARILY BATCHED FORWARD CALLS. LATENCY IS REPORTED SECONDS PER WINDOW ON ONE MI300X.

Configuration	NFE	η	ME	CRPS↓	WS↓	MAE	RMSE	PICP	Width	Latency
DDPM	1000	1	10000	0.982	8.071	1.385	2.828	0.875	4.535	21.17
DDIM, selected	10	0.5	100	0.886	9.119	1.156	2.382	0.648	2.895	0.21
DDIM, $\eta = 0$	10	0	100	0.915	9.242	1.266	2.567	0.664	2.832	0.21
Smart persistence	–	–	–	1.578	–	1.578	3.264	–	–	–
Persistence	–	–	–	1.808	–	1.808	3.300	–	–	–

VI. DISCUSSION

The evidence supports a configuration-specific speed–coverage trade-off. It does not show that short schedules necessarily collapse uncertainty. Intermediate step counts were not tested, attention offsets were not controlled separately, and the clipping cap may affect the upper tail. Finite-M scoring can also favor configurations differently from latent-distribution scoring. A causal explanation requires a factorial comparison of step count, η, attention randomness, and clipping under matched initial noise, followed by repeated-seed evaluation.

The generated sky images do not reproduce observed clouds convincingly in Fig. 1. Their role here is an auxiliary joint output, not a validated visual forecast. Adding temporal modeling is a reasonable hypothesis, but its benefit must be measured rather than assumed. Similarly, history masking is not a substitute for training strong unimodal comparators.

For interval-based use, larger ensembles, explicit quantile conventions, and held-out recalibration are necessary checks. Conformal adjustment is a possible route, but ordinary exchangeability-based guarantees cannot be assumed for overlapping temporal windows or distribution shift [16]. No calibrated interval results or reserve-sizing experiment are reported here. The study is further limited to one site, one training run, one sampling seed, and 20 test days. Exact split manifests, timestep indices, normalization and clipping code, bootstrap weighting, random-number streams, and synchronized timing settings are needed for a fully reproducible follow-up.

VII. CONCLUSION

A matched-checkpoint comparison shows that ten-step DDIM can substantially reduce sampling time for this multimodal PV forecaster. On the evaluated subset, it improves endpoint empirical CRPS while reducing empirical interval coverage from 87.5% to 64.8%. The full-test DDIM result is 1.144 kW CRPS at the endpoint under the stated timestamp convention. Fast sampling is promising, but nominal calibration and a schedule-length mechanism remain unestablished. Controlled schedule ablations, finite-ensemble-aware verification, and held-out recalibration are the next priorities.

ACKNOWLEDGMENT

The authors thank the creators of SKIPP'D for making the dataset available.

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